



PIEZOELECTRIC WALKWAY AT NORTHEASTERN UNIVERSITY

BY: TEAM 2 - ENVIRO-MENTALISTS

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PROJECT PURPOSE AND GOAL

Problem:

- Wastage of kinetic energy generated by pedestrian movement on campus
- Rising energy costs at the university.
- Increasing demand for sustainable energy solutions.

Purpose:

- Reduce reliance on conventional energy sources and promote sustainability on campus.
- Transform wasted kinetic energy from foot traffic into usable power by installing a piezoelectric walkway at Northeastern University, contributing to sustainability.

Goal:

The goal of our project is to minimize the carbon footprint, reduce electricity costs and promote sustainability. By doing so, we aim to generate clean, renewable energy through the movement of pedestrians, supporting the university's commitment to sustainability and furthering its green energy initiatives.

PROJECT PURPOSE & GOAL

Objectives:

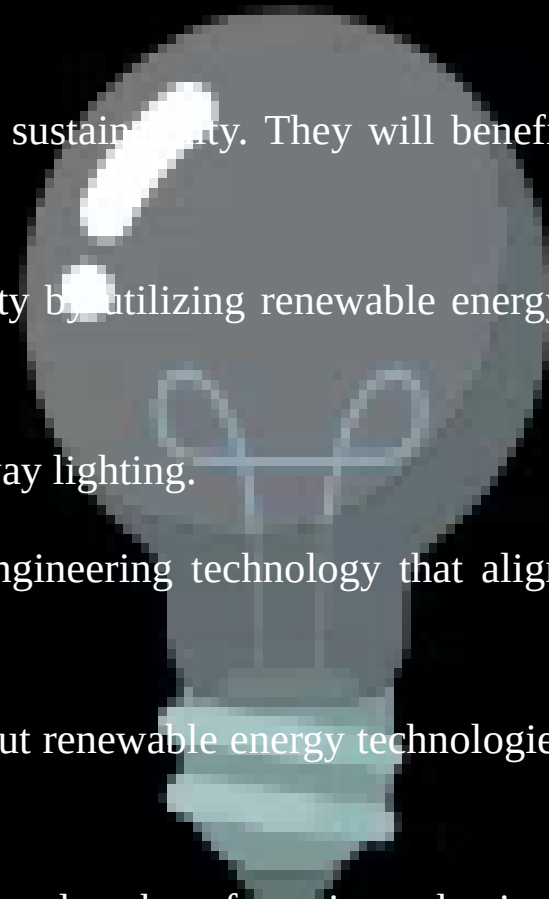
1. **Energy Conversion:** Convert at least 75% of the kinetic energy from foot traffic into usable electrical energy.
2. **Lighting Effectiveness:** Successfully power 70% or more of the light poles along the designated pathway using the energy generated by the tiles.
3. **System Durability:** Ensure the piezoelectric tiles maintain functionality under daily traffic of 8,000+ pedestrians.
4. **Cost Savings:** Achieve a 20% reduction in electricity costs for pathway lighting within the first year of implementation.
5. **Energy Storage:** Install an energy storage system capable of storing at least 25 kWh of electricity for continuous lighting during off-peak hours.
6. **Performance Validation:** Conduct a performance study to evaluate the feasibility of scaling the system to other pathways or campus areas.

PROJECT PURPOSE & GOAL

Benefit to the Client:

The client is Northeastern University, an institution dedicated to innovation and sustainability. They will benefit from this project by:

1. **Reduction in Energy Consumption:** Decrease dependency on grid electricity by utilizing renewable energy generated from footsteps, leading to energy savings.
2. **Cost Savings:** Lower electricity costs by using self-generated power for pathway lighting.
3. **Practical Sustainability Solution:** Demonstrate a feasible application of engineering technology that aligns with the university's sustainability goals without disrupting campus operations.
4. **Student Engagement:** Offer an educational platform for students to learn about renewable energy technologies, fostering engagement with sustainability and engineering practices.
5. **Climate Action Goals:** Contribute to the university's climate action plans and carbon footprint reduction targets by integrating renewable energy into its infrastructure.



PROJECT PURPOSE & GOAL

Why is there hope that we will solve the problem?

We are confident in the success of our project because it is based on a practical, innovative solution that aligns with Northeastern University's goals for sustainability. By utilizing piezoelectric tiles, we are turning everyday foot traffic into a valuable resource, which is both efficient and cost-effective for the university. We have carefully considered the walkway's high foot traffic and chosen an energy solution that is easy to implement and maintain. With our team's strong engineering background and commitment to sustainable practices, we believe our approach will effectively address the problem and contribute to the university's long-term energy goals. Several other key factors are:

- **Proven Technology:** Piezoelectric energy generation has been successfully implemented in various locations worldwide, demonstrating its reliability and effectiveness..
- **Cost Reduction:** By generating energy from foot traffic, we provide a solution that will significantly reduce electricity costs for the university.
- **Environmental Impact:** The project will help decrease the university's carbon footprint, aligning with its environmental goals.

TECHNICAL OVERVIEW

APPROACH

Site Selection

- Identify the high-foot-traffic path (e.g., Krentzman Quad to Snell Library).

Tile Installation

- Install Pavegen V3 piezoelectric tiles along the pathway.

Energy Capture

- Pedestrian footsteps generate energy via piezoelectric tiles.

Energy Storage

- Energy is collected through copper wiring and stored in lithium-ion batteries.

Energy Conversion

- Inverters convert DC to AC for LED lighting.

Lighting Integration

- Power energy-efficient LED lights along the walkway.

Data Monitoring

- Real-time foot traffic sensors monitor and optimize system performance.

Testing & Commissioning

- Verify system functionality, including energy generation, storage, and lighting.

Pavegen: How a footstep's energy is converted to electrical power

TECHNICAL OVERVIEW

What will be Produced? And How will it be Produced?

Piezoelectric Technology: Piezoelectric technology generates electricity from pressure or vibrations. In this project, the Pavegen V3 tiles use foot traffic to create electrical energy. When people walk over the tiles, they compress slightly, and the piezoelectric materials inside convert this pressure into electricity.

Pavegen V3 Tiles: The Pavegen V3 tiles are designed to capture energy from footsteps. The Pavegen V3 tiles used in this project are triangular and equipped with electromagnetic generators that produce power with each step. These tiles are durable, slip-resistant, and designed for high foot traffic. Each tile generates electricity when pressure is applied, making them perfect for public walkways like the one from Krentzman Quadrangle to Snell Library.

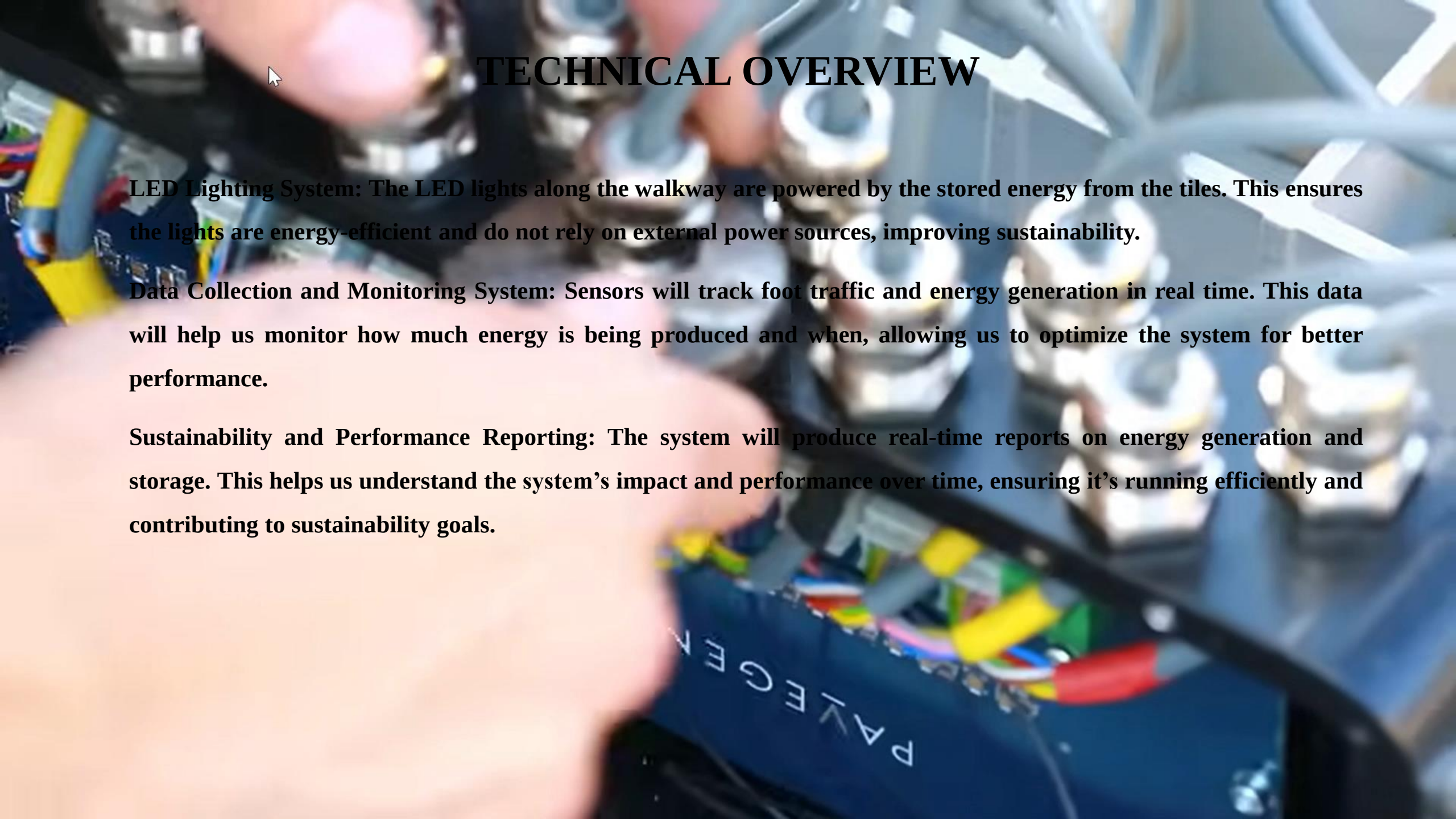
Energy Storage System: The generated electricity is transmitted via copper wiring to a junction box, then stored in lithium-ion batteries. These batteries store energy during high traffic periods (like the day) and power the walkway lights during low traffic times (like at night).



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Scroll for details





TECHNICAL OVERVIEW

LED Lighting System: The LED lights along the walkway are powered by the stored energy from the tiles. This ensures the lights are energy-efficient and do not rely on external power sources, improving sustainability.

Data Collection and Monitoring System: Sensors will track foot traffic and energy generation in real time. This data will help us monitor how much energy is being produced and when, allowing us to optimize the system for better performance.

Sustainability and Performance Reporting: The system will produce real-time reports on energy generation and storage. This helps us understand the system's impact and performance over time, ensuring it's running efficiently and contributing to sustainability goals.

IMPLEMENTATION PLAN

- 1. Site Preparation and Planning:** We begin by conducting an on-site survey of the walkway between Krentzman Quad and Snell Library. This will involve assessing foot traffic patterns and ensuring compatibility with the existing lighting system. The survey helps identify optimal tile locations and potential installation challenges, allowing us to plan for a smooth process.
- 2. Design Phase:** Following the site survey, the design phase focuses on the final layout of the piezoelectric tiles. The team will ensure the tiles are placed for maximum energy generation without compromising walkway safety. Electrical engineers will design the connections to integrate the tiles with the campus's lighting system, including energy storage and power transmission.
- 3. Pre-Installation Phase:** Before installation begins, we will implement traffic diversion and site preparation measures to ensure pedestrian safety and minimize disruptions. Temporary pathways and barricades will be installed to guide pedestrians around the construction area. Communication with campus stakeholders will be maintained to ensure the university community stays informed about the project.
- 4. Excavation Phase:** During the excavation phase, we will prepare the site by digging and leveling the ground to ensure a stable base for the piezoelectric tiles. This step is crucial for proper tile alignment and ensuring the walkway's durability and smooth installation.

IMPLEMENTATION PLAN

THE TILE CONSISTS OF
THREE COMPONENTS

5. **Tile Installation:** With the electrical infrastructure in place, we'll install the piezoelectric tiles along the walkway. The tiles will be securely fastened to the base to ensure they remain intact despite heavy foot traffic. The installation will be conducted based on optimal locations identified from the foot traffic survey.

AN ELECTRO-MAGNETIC
GENERATOR

6. **Electrical Connections and Wiring:** In this phase, we'll establish the electrical infrastructure by connecting the tiles to an underground junction box and linking them to lithium-ion battery storage. The wiring will be done using durable copper cables to ensure long-term reliability and resistance to environmental conditions.

A SINGLE
TRIANGULAR
COMPOSITE TILE

7. **Testing Phase:** Once the tiles are installed, we will begin system testing. This will include simulating foot traffic to measure the energy generated and ensure the power is routed correctly to the lighting system. Engineers will also test the storage system to ensure efficiency and durability under various environmental conditions.

AND **YOU!**

8. **Post-Installation Phase:** After successful installation and testing, we'll provide the university with detailed documentation and maintenance guidelines. Training sessions will be held for the university's maintenance staff to ensure proper system operation. A real-time monitoring dashboard will be set up to allow the university to track energy harvest and assess system performance over time.



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IMPLEMENTATION PLAN

9. **Work Breakdown Structure (WBS):** For our Piezoelectric Walkways project, we will follow a structured Work Breakdown Structure (WBS). In the Initiation phase, we will define the project scope, establish the project team, and engage key stakeholders such as university departments and regulatory bodies. During Planning, we will conduct a feasibility study, ensure regulatory compliance, and create detailed designs for the walkway, including wiring and drainage plans. In the Execution phase, we will prepare the site, install the piezoelectric tiles, wire the system, and perform thorough quality control and testing. Finally, in Closure, we will review the project deliverables, establish a maintenance plan, and complete the final project report. This approach will help us stay on track and ensure the project's successful completion.
10. **Schedule:** Our project schedule spans 79 days and ensures timely delivery through well-defined timelines for each phase. We've allocated 9 days for initiation to establish project foundations, followed by 31 days for detailed planning, where designs, feasibility, and procurement preparations are finalized. The execution phase is the longest, at 33 days, carefully scheduled for construction, installation, and system integration tasks. Finally, we've dedicated 4 days for closure activities and 2 days for compiling the final project report, ensuring a comprehensive and organized completion.

IMPLEMENTATION PLAN

10. RACI Matrix: Our RACI matrix outlines the distribution of responsibilities for each task in the project. It designates specific roles to ensure clear accountability and efficient collaboration. For example, the **Project Manager** is **Responsible** for key tasks like the **Kick-Off Meeting** and **Defining the Project Scope** while also being **Accountable** for their completion. Other stakeholders, like the **Renewable Energy Development Manager**, **Civil Engineer**, and **Procurement Manager**, are **Consulted** and **Informed** depending on the task's requirements. This ensures that everyone involved in the project understands their specific role, which helps streamline decision-making, communication, and task execution throughout all project phases. The matrix is an essential tool for keeping the project on track and ensuring alignment among team members and stakeholders.

FINANCIAL PLAN

The total budget for our project, which includes the installation of piezoelectric tiles along the walkway between Krentzman Quad and Snell Library, amounts to **\$1,388,836.69**. The budget was calculated considering the following costs:

- **Labor Costs (\$715,444.20):** Covers all wages for workers, engineers, and electricians needed for installation and setup of the system.
- **Equipment Costs (\$833,432.00):** Includes the purchase and installation of piezoelectric tiles, electrical infrastructure (wiring, junction boxes), and energy storage systems.
- **Miscellaneous Costs (\$40,000):** Allocated for unforeseen expenses like extra tools, permits, and minor purchases during installation.
- **Contingency (7.41%):** \$117,735.73 reserved for any unexpected issues such as price increases or delays.

Final Project Cost: \$1,388,836.69 after rebates, covering the entire project from planning to completion.

FINANCIAL PLAN

To assess the financial viability of the project, we will calculate the Present Worth (PW) using the net savings from energy generation and maintenance costs over the project's lifespan.

The assumptions made to calculate the financial viability of our project are:

Annual Savings: \$80,000 from energy savings.

Maintenance Costs: \$20,000 annually for upkeep.

Project Life (n) : 20 years of operation.

Discount Rate (r) : 5% to account for time value of money.

Net Savings = Annual Savings - Annual Maintenance Costs = \$80,000 - \$20,000 = \$60,000

Thus, Present Worth (PW) = Initial Investment + Net Savings $\times [1 - (1+r)^{-n} / r]$, where: Initial Investment = \$1,388,836.69

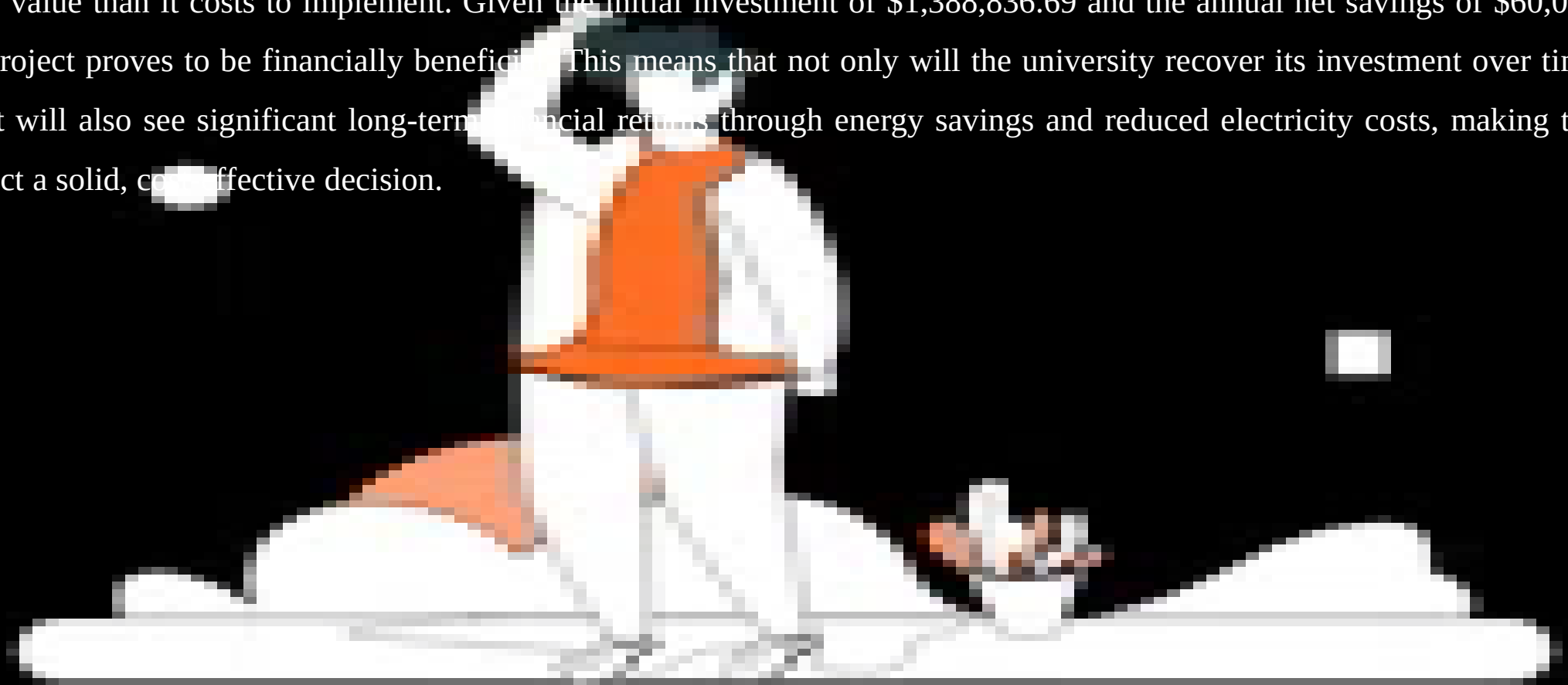
Substituting the values into the formula:

PW = 1,388,836.69 + 60,000 $\times$ 12.4622

PW = \$2,136,568.69

FINANCIAL PLAN

The Present Worth (PW) of the project is \$2,136,568.69, which shows that over the 20-year lifespan, the project will generate more value than it costs to implement. Given the initial investment of \$1,388,836.69 and the annual net savings of \$60,000, the project proves to be financially beneficial. This means that not only will the university recover its investment over time, but it will also see significant long-term financial returns through energy savings and reduced electricity costs, making this project a solid, cost-effective decision.



PIEZOELECTRIC WALKWAY AT NORTHEASTERN UNIVERSITY

PROJECT OVERVIEW

- **Project Purpose and Problem:** The purpose of this project is to install piezoelectric tiles on the walkway between Krentzman Quad and Snell Library, harnessing kinetic energy from foot traffic to generate power for campus lighting. The problem addressed is the university's increasing energy costs and the need for sustainable energy solutions.
- **Project Goal:** The goal is to reduce electricity costs and promote sustainability by generating clean energy through the movement of pedestrians, while also enhancing the campus's commitment to green energy initiatives.
- **Benefit to the Client:** The university will benefit from reduced energy bills, an environmentally friendly power generation system, and a long-term energy-saving solution. Additionally, this project will enhance the campus's reputation as a leader in sustainable initiatives.
- **Impact if Funded:** If funded, the project will provide long-term savings of approximately \$60,000 annually, reduce carbon footprint, and contribute to a greener, more energy-efficient campus environment. The university will see a significant return on investment, with financial and environmental benefits extending for the next 20 years.





THANK YOU